TRUNKBELAY · STEM CHECK

TrunkBelay STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: S30 to plausible pending the TRL 4 test to failure (O7, option C); S31 closing route with the 20 deg lean limit and drill rule (O8); 19 within, 13 plausible for option A
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.9.1 upgrade) of the TRL 3 design (option A) and of concept options B and C; two detail failures fixed under the standing instruction (TKB-DEC-001 D-S1 from the D3 model, D-S2 from the independent review); the strength margin (S30) proposed to Amish; independent review recorded

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Verdict: the working principle of the current design (option A, a rigid V frame whose grip rises with the load through its lever arm) sits within known physics, and so do its strength (after one fix), lowering, heat, fit and tether claims. Of the 32 claims for option A, **19 are within known science and 13 are plausible, to confirm at TRL 4**. The plausible rows are the friction of rubber and sleeved chain on wet coconut bark (S3, the claim the lock rests on), the slack drop (S2), settlement with pad shear (S5), wear of the slot faces (S10), the rigging time and the time to the ground (S17, S23), the bark under the pads' lower edge on green wood (S19, S21), the victim turning head up (S22), corrosion and ageing (S25, S26) and a leaning trunk (S31). The row that failed is the strength margin: the frame's estimated breaking load is about 5.0 kN, 5.1:1 on the one-person load, and about 4.5:1 if the chain's whole pull falls on one slot, below the 5:1 to 10:1 that rope rescue teams aim for (S30); raising it costs carried mass, a safety trade-off put to Amish (TKB-DEC-001 O7). Amish decided on 2026-10-25 ("i concur with your recommendations") to keep the 2.5 kN design load, raise the TRL 4 proof load to 3.5 kN and decide after a test to failure of a sample frame, with the stronger frame as the fallback below 5:1, so S30 is now plausible, to confirm by that test. For the leaning trunk (S31) he set a 20 deg lean limit in PR-003 and a drill rule for the frame's side (O8). `stem_check` is **open**. Two details failed and were fixed under Amish's standing instruction of 2026-10-23: the tether quick link did not pass a 6 mm chain link (D-S1, S27, S28), and the independent review showed that the 35 mm cleat web failed if the chain's pull fell on one slot, so the spine top is raised 20 mm to a 55 mm web (D-S2, S9). Options B and C have no failing row; each rests on a plausible claim that only a test could settle. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the calculation note's script (`docs/04-calcs/sizing.py`, TKB-CAL-001) and so the model.

> **Safety:** TrunkBelay holds and lowers a person at height. A verdict of "within known science" means the claim is consistent with physics on paper, not that the kit is safe. Nobody is lowered with it before the collar has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height (TKB-BLD-001, section 6).

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` by `docs/04-calcs/stem_check.py`).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Collar carries about 0.99 kN holding a 100 kg person with the victim set	Mathematics	Units and dimensional analysis	Weight = m g (100 kg person, 0.96 kg of triangle, sling and carabiner)	claimed 0.99 kN, estimate 0.99 kN (ratio 1.00)	Within known science	TKB-CAL-001 [A1]
S2	A drop of up to 30 mm onto the rope when the feet are freed stays inside the frame's strength margin	Physics	Mechanics and statics	Energy balance of a rigid mass on 1 m of rope, F = W (1 + sqrt(1 + 2 k h / W)); 4 % stretch (CAL) and the stiffest plausible rope, 2 %; frame limit 2.5 kN x 1.36 (the bars' combined factor, [C1b])	30 mm of slack: 2.6 kN (4 %) to 3.0 kN (2 %), above the 2.5 kN of R1 but inside 3.4 kN; 100 mm with 2 % rope: 4.3 kN, beyond it. The 30 mm is a drill limit, not physics (independent review); to close: the drill caps slack and the TRL 4 proof load is raised above 3 kN (TKB-DEC-001 O7)	Plausible, to confirm at TRL 3	
S3	The collar locks on wet bark: combined friction of pads and sleeved chain at least 0.58 on a 200 mm trunk (0.9 assumed)	Physics	Mechanics and statics	Self-locking couple: lock if (mu_pad + mu_chain) >= h / e (TKB-CAL-001 section 3); friction of rubber and nylon on wet coconut bark is not published	needed 0.58, assumed 0.9 (0.5 + 0.4), locking factor 1.54; no measured value; to close: friction on wet bark measured at TRL 4 (BLD-001 first check 1)	Plausible, to confirm at TRL 3	
S4	The simplified lock condition h / e in the CAL is on the safe side of the full equilibrium	Mathematics	Numerical method	Moments about the trunk axis with the pad contacts at r cos 30 deg, the chain at R_c on the far side and the two pad normals (friction mu N / cos	1.539 <= 1.63	Within known science	TKB-STM-001 S4 (full equilibrium gives 1.63 against 1.54)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				30 deg); same frictions as S3			
S5	The frame settles less than the 20 mm of R1 at 2.5 kN	Physics	Strength of materials	Pad squash at a 6 MPa compression modulus plus one link of chain seating, levered by e / h (TKB-CAL-001 [B5]); the independent review adds shear of the pads under their share of the vertical friction (G about 1.1 MPa) and tilt of the chain at the back	10 mm in the CAL; about 13 mm with pad shear, 14 to 18 mm with chain tilt, against 20 mm; to close: settlement measured in the TRL 4 proof test	Plausible, to confirm at TRL 3	
S6	Bearing bars stay below the welded (heat-affected) proof strength of 6082-T6 at 2.5 kN	Physics	Strength of materials	RHS 50 x 40 x 3 from the spine: bending in plan, vertical bending from the pad's friction and torsion from the bottom-edge contact, combined by von Mises (independent review); f0,haz = 125 MPa (EN 1999-1-1 values for welded 6082-T6)	92 MPa <= 125 MPa	Within known science	TKB-CAL-001 [C1b] (factor 1.36; 1.60 in plan bending alone)
S7	Bar to spine fillet welds stay below the weld metal strength	Physics	Strength of materials	Weld group in bending, 5 mm fillets (3.5 mm throat) all round; ER5356 weld metal 210 MPa; about 57 MPa equivalent with the shear (independent review), still factor 3.7	40 MPa <= 210 MPa	Within known science	TKB-CAL-001 [C2] (factor 5.3)
S8	The load eye does not	Physics	Strength of materials	Two shear planes through the 13 mm	2.5 kN <= 38.9 kN		TKB-CAL-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	tear out at 2.5 kN			eye stack (spine and doubler rings), 14 mm of edge beyond the 22 mm eye, 0.577 x 185 MPa HAZ ultimate		Within known science	[C3] (39 kN, 16 times)
S9	The 5 mm cleat web beside a slot stays below the welded proof strength with all of the chain's pull on one slot (450 mm trunk)	Physics	Strength of materials	Ligament in bending beside the slot, web 55 mm deep (D-S2); the two legs need not share the tension (capstan ratio about 6 round the wrap), so the worst case puts it all on one slot; with the old 35 mm web this case gave 162 MPa (factor 0.77) and failed; f0,haz = 125 MPa	103 MPa <= 125 MPa	Within known science	TKB-DEC-001 D-S2; TKB-CAL-001 [C4b] (factor 1.21; 2.43 shared equally)
S10	Steel chain links bearing on the aluminium slot faces do not wear or dent the cleat in service	Engineering and technology	Manufacturing tolerances	Projected bearing of a 6 mm link on the 5 mm web against about 1.5 x f0,haz; local line contact and repeated use make wear the open question	projected bearing 92 MPa against about 190 MPa; wear and denting not calculable on paper; to close: inspect the slot faces after the TRL 4 proof tests (TKB-DEC-001 item 11)	Plausible, to confirm at TRL 3	
S11	The 6 mm grade 80 chain holds the largest chain tension with a wide margin	Physics	Strength of materials	Grade 80, 6 mm: minimum breaking force 45.2 kN, working load limit 1.12 t (S3i, EN 818-2 class); checked with all of the pull in one leg (independent review); a link bearing on a flat web is not the link-on-link case of the rating, a reason the factor is kept large	11.1 kN <= 45.2 kN	Within known science	TKB-CAL-001 [C5] (factor 4.1 in one leg, 8.1 shared)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S12	The rope at its figure-eight loop holds 2.5 kN wet, with margin	Chemistry	Materials and polymer chemistry	Basis here: EN 1891 type A holds 15 kN for 3 minutes with figure-eight terminations, less 15 % wet (the CAL's [C6] uses 22 kN x 0.65 knot efficiency, 14.3 kN dry); new-rope values, so abrasion and descender bends are left to the retirement rules (S26)	2.5 kN <= 12.8 kN	Within known science	TKB-CAL-001 [C6] (14 kN dry at the knot)
S13	Brake hand force for a 100 kg person stays under the 150 N of R5	Physics	Mechanics and statics	Capstan friction T1/T2 = exp(mu theta): rope on aluminium mu 0.2 over 540 deg in the descender and 180 deg over the brake carabiner; cross-check: EN 341 limits the holding force to 200 N at the device's maximum rated load (150 kg for the cited class), about 135 N at 101 kg, 72 N with the brake carabiner. Within for a certified device on dry rope: the capstan estimate reaches 150 N at a friction of 0.15 and 282 N at 0.10 (independent review)	80 N <= 150 N	Within known science	TKB-CAL-001 [D1]
S14	Lowering 100 kg through 25 m turns 24.8	Physics	Thermodynamics	Energy balance m g h = heat in the descender, brake carabiner and	in 24.75 kJ, out 24.76 kJ	Within known science	TKB-CAL-001 [D2]

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	kJ into heat; the descender warms by 44 K at most			hand; dT = Q / (m c), 0.53 kg of aluminium, c 900 J/(kg K): bulk rise 44 K, the body near 79 C from 35 C ambient, hotter at the rope contact; the hand holds only the handle and the brake strand (S29); EN 341 class A allows 7.5 x 10^6 J of descent energy			
S15	30 m of rope lowers from 25 m	Mathematics	Orders of magnitude	Rope used = 25 m drop + 1 m to the person + 1.5 m of knots + 0.5 m round the brake carabiner	28 m <= 30 m	Within known science	TKB-CAL-001 [E1]
S16	The 110-link chain fits every trunk from 200 to 450 mm and leaves at least 4 links in the adjustable tail	Mathematics	Numerical method	Chain path round the trunk from the model: loop links + fixed end + tail = 110; links counted on the 450 mm trunk; on a 200 mm trunk 59 links (about 1.1 m) hang free and are stowed in the bag's clip loop	15 links >= 4 links	Within known science	TKB-CAL-001 [E2]; cad/src/model.py checks
S17	A trained climber fixes the collar and clips the victim in under 5 minutes once in position	Biology	Ergonomics	Task time estimate summed step by step (TKB-CAL-001 [E3]); work at height on a climbing device, hot and possibly wet	estimate 4.5 min against 5 min; a thin margin, no timed drill yet; to close: timed drill at 3 m at TRL 4 (BLD-001 first checks; PR-004)	Plausible, to confirm at TRL 3	
S18	The rescuer carries under 5 kg up the trunk	Engineering and technology	Mass budget	Sum of the carried items from the model and catalogue masses,	4.67 kg <= 5 kg	Within known science	TKB-CAL-001 [F2]

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				with the cited descender at 600 g (Petzl I'D S class) and the 55 mm cleat web of D-S2			
S19	The rubber pads do not crush the bark at 2.5 kN	Biology	Plant and animal biology	Pad pressure from the CAL against the compressive strength of coconut stem wood, 9.28 MPa air dry at 400 kg/m3 average density (Rana et al. 2015; grain direction as reported); green or wet wood is typically 40 to 50 % weaker, and a cocked pad bears on a strip near its lower edge (independent review)	1.9 MPa mean, about 3.8 MPa at the lower edge, against about 4.6 to 5.6 MPa green; no margin left at the edge; to close: crush test on green stem sections at TRL 4	Plausible, to confirm at TRL 3	
S20	Bare chain links outside the sleeve mark the bark less than the 5 mm of R8 (static)	Physics	Strength of materials	Line load T / R = 23 N/mm on a 6 mm wire; the wire sinks until the contact width w = q / 9.28 MPa (2.5 mm); depth = r - sqrt(r^2 - (w/2)^2); static only: dragging while tightening is not included; about 1.4 mm on green wood at 4.6 MPa	0.27 mm <= 5 mm	Within known science	TKB-CAL-001 [G1] (bare links may mark, R8)
S21	The collar does not penetrate the stem and leaves no mark deeper than 5 mm	Biology	Plant and animal biology	Palms have no vascular cambium, so any mark is permanent (Broschat 2019): the shallow static marks of S20 stay; the collar has no spikes, screws or cleats; pad	no penetrating part; marks about 0.3 to 1.3 mm static; to close: bark inspection after the TRL 4 lowerings (PR-008)	Plausible, to confirm at TRL 3	

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				crushing depends on S19 (reworded after the independent review)			
S22	The victim turns head up in the triangle once the feet are freed	Biology	Human physiology and biomechanics	Statics: a hanging body settles with its centre of mass below the attachment; the triangle and chest sling attach at the hips and chest, above the centre of mass when head up	attachment above the centre of mass when head up, but hanging head down from the hips is also a stable state; that the body passes from one to the other unaided is not shown by statics (independent review); to close: dummy drill at TRL 4	Plausible, to confirm at TRL 3	
S23	The kit brings a stranded climber down in minutes, against the 30 to 150 minutes reported	Mathematics	Orders of magnitude	Rigging (S17) + lowering 25 m at 0.3 m/s + about 3 minutes to climb to the victim (not timed); reported waits from the PRB sources	about 9 min against 30 to 150 min; rests on S17 and an untimed climb; to close: timed drill at TRL 4	Plausible, to confirm at TRL 3	
S24	The upward pull of R2 does not arise when the collar is at least 300 mm above the victim's attachment	Physics	Mechanics and statics	The load strand runs from the descender down to the attachment point below it, so the force on the load eye points down; the upward case needs the attachment above the collar (the drill rule, decision 16A). A victim 0.3 m to the side on 1 m of rope adds about 0.3 W sideways, not counted in the	-1 (vertical component of the rope pull, up positive, per unit load) <= 0 (vertical component of the rope pull, up positive, per unit load)	Within known science	TKB-DDR-003 16A; R2 stays at risk on the drill, not on the physics

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				lock analysis (independent review)			
S25	Bare 6082-T6 frame in contact with steel chain, zinc-plated keeper and stainless screws survives humid coastal use	Chemistry	Electrochemistry and batteries	Aluminium is anodic to steel and stainless steel in a wet joint; the pad screws sit in stainless rivet nuts with barrier paste, the chain touches the slots only while rigged, and the kit is stored dry in its bag	galvanic attack needs a standing electrolyte; the controls are in the build plan; the rate on bare 6082 in sea air is not calculated; to close: inspect the frame after the TRL 4 trial season (TKB-RSK-001 FMEA)	Plausible, to confirm at TRL 3	
S26	Rope, webbing and rubber pads keep their properties in tropical sun and heat for their service life	Chemistry	Materials and polymer chemistry	Polyamide and natural rubber degrade in UV and ozone; the kit is stored in its bag out of the sun, and rope, slings and pads are retired on the makers' inspection rules	time-dependent; controlled by storage and inspection, not by a margin; to close: makers' retirement rules adopted in the drill card at TRL 4 (TKB-RSK-001)	Plausible, to confirm at TRL 3	
S27	The tether link threads the chain's end link: wire diameter inside the 6 mm grade 80 link's 7.8 mm opening	Engineering and technology	Manufacturing tolerances	Inner width of 6 mm grade 80 chain at least 7.8 mm (S3i, EN 818-2 class); the BOM's 8 mm quick link failed this check and was replaced by a 6 mm grade 80 connecting link (7 mm at the chain end) through a 10 mm tether hole	7 mm <= 7.3 mm	Within known science	TKB-DEC-001 D-S1 (fix under the standing instruction)
S28	The brake carabiner has the 22 mm spare eye to itself, clear of the tether	Engineering and technology	Manufacturing tolerances	D3 model interference check (cad/fidelity.json) after D-S1: the tether link and the fixed end of the chain sit in front of the spine, the	0 mm3 of overlap <= 5 mm3 of overlap	Within known science	TKB-DEC-001 D-S1; cad/fidelity.json

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				brake carabiner alone in the spare eye			
S29	The brake strand the rescuer holds stays far below the 48 C touch limit of EN 341	Physics	Heat transfer	8 % of the 24.8 kJ goes into the hand and the brake strand running through it; spread over the 1.95 kg of rope that passes the hand (c about 1.7 kJ/(kg K)) it warms the rope by about this much; limit: 48 C less 35 C ambient. The descender body (85 %, up to 44 K) is held only by its handle	0.6 K <= 13 K	Within known science	TKB-CAL-001 [D2]
S30	The collar's static system safety factor on a one-person load sits within the 5:1 to 10:1 that rope rescue teams aim for	Engineering and technology	Standards practice	Breaking load estimated from the weakest frame member at the welded ultimate strength, over the 0.99 kN working load; rescue practice ratios 5:1 to 10:1, 15:1 once common (Piatt 2023); EN 795 tests anchor devices statically at 12 kN	5.0 kN with the chain's pull shared (5.1:1, the bars govern), 4.5 kN with all of it on one slot (4.5:1): below 5:1 in the worst case. Amish decided on 2026-10-25 (TKB-DEC-001 O7, option C): keep the 2.5 kN design load, raise the TRL 4 proof load to 3.5 kN and decide after a test to failure of a sample frame, with the stronger frame (option B) as the recorded fallback if it shows less than 5:1	Plausible, to confirm at TRL 3	Test to failure of a sample frame and the 3.5 kN proof test, TRL 4 (TKB-DEC-001 O7)
S31	The collar locks on a trunk leaning up to	Physics	Mechanics and statics	Self-locking condition on a leaning trunk, frame on the	locking factor 1.30 at 15 deg, 1.21 at 20 deg, 1.02 at 30 deg on	Plausible, to confirm at TRL 3	

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	20 deg with the frame on either side			adverse side: need = h cos t / (e cos t - h sin t) (independent review), 200 mm trunk, the same assumed friction 0.9	the adverse side; Amish decided on 2026-10-25 (TKB-DEC-001 O8, option C): a 20 deg lean limit in PR-003 and a drill rule that sets the frame on the favourable side of a leaning trunk; to close: friction test and the drill on a leaning palm		
S32	The tether hole holds the fixed end's share of the chain tension if the keeper is unseated	Physics	Strength of materials	Two shear planes of the 7 mm web strip beside the 10 mm tether hole, 5 mm thick, 0.577 x 185 MPa HAZ ultimate, against the chain tension at the R1 load (2.2 kN at the working load); the tether is not a working load path	5.5 kN <= 7.5 kN	Within known science	TKB-DEC-001 D-S1 (factor 1.4 at the R1 load, 3.4 at the working load)
B1	Option B: a strap wrapped 4 turns and tied off as a Klemheist holds 2.5 kN along a vertical 300 mm trunk	Physics	Mechanics and statics	Capstan friction over the wrapped length: axial holding about 2 pi n mu T with the band tension T at least W / 2 in each wrap (mu 0.3 assumed, polyester on wet bark); friction hitches round large trunks under axial load are not covered by a cited source	holding about 3.8 times the load on paper; the self-tightening of the hitch round a large, wet, tapering trunk is the open question; not carried forward (option B is not carried forward unless Amish chooses it)	Plausible, to confirm at TRL 3	
B2	Option B: the 45 mm polyester anchor strap	Physics	Strength of materials	EN 795 anchor devices are tested statically at 12 kN (metallic) or 18 kN (textile, no service-	2.5 kN <= 18 kN	Within known science	Catalogue class; TKB-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	holds 2.5 kN with margin			life marking) for one user (EN 795 summary, Vandernet)			
B3	Option B: the wraps press the bark far below its crushing strength	Biology	Plant and animal biology	Band pressure p = T / (R w) at T = W / 2 on a 300 mm trunk, 45 mm webbing, against 9.28 MPa (Rana et al. 2015)	0.19 MPa <= 9.28 MPa	Within known science	TKB-OPT-001
B4	Option B: a climber ties the 4-turn wrap at height in about 90 s	Biology	Ergonomics	Task time estimate: four wraps and a Klemheist with a 4 m strap, one hand on the trunk	estimate 90 s against 60 s for the collar; not timed; not carried forward (option B)	Plausible, to confirm at TRL 3	
C1	Option C: four M10 x 120 coach screws hold 2.5 kN in the stem	Physics	Strength of materials	Johansen steel-plate-to-timber estimate per screw with timber formulas (EN 1995-1-1 form) at 400 kg/m3: 8.6 kN each; palm stem wood is fibre bundles in soft tissue, not timber, so the embedment strength is unknown	about 35 kN for four screws on timber formulas against 2.5 kN; not valid for palm wood without a test; not carried forward (option C is not carried forward unless Amish chooses it)	Plausible, to confirm at TRL 3	
C2	Option C: four 10 mm holes 110 mm deep do not seriously harm the palm	Biology	Plant and animal biology	Palm stem wounds are permanent (Broschat 2019); every use leaves four more holes, an entry for pests and rot; R8 limits marks to 5 mm	the holes are permanent and fail R8 by design; harm to the palm is not quantified; not carried forward (option C); fails PR-008 by design	Plausible, to confirm at TRL 3	
C3	Option C: the 10 mm steel eye lug does not tear out at 2.5 kN	Physics	Strength of materials	Two shear planes of the 10 mm lug below the 22 mm hole (39 mm of material), 0.577 x	2.5 kN <= 185 kN	Within known science	TKB-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				410 MPa ultimate (S275 class)			
C4	Option C: driving four coach screws by hand at height fits the 5 minutes of R4	Biology	Ergonomics	Pilot drilling and driving 4 x 120 mm screws with a ratchet while hanging on a climbing device, estimate 2 min a screw	estimate about 8 min for the screws alone against 5 min for the whole rig; not carried forward (option C); at risk against PR-004	Plausible, to confirm at TRL 3	

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3** (here: to confirm at TRL 4, since the repo is at TRL 3 and the open rows need parts or trunks), **Fails, fix proposed**, **Fails**. An empty "Closed by" cell is an open row; what closes it is in the check column.

Fixes made under the standing instruction

Table 2. Detail failures and their fixes.

Item	What failed	Fix (TKB-DEC-001)	Result
Fixed-end tether	Found by the detailed (D3) model: the 8 mm quick link (BOM 14) cannot pass through the end link of 6 mm grade 80 chain, whose inner width is at least 7.8 mm. Placed in the spare eye beside the brake carabiner, the chain's hooked end link could not lie anywhere without striking the carabiner or the doubler rings	D-S1: the fixed end is tethered with a 6 mm grade 80 connecting link (7 mm at the chain end) through a new 10 mm tether hole 12 mm behind the spine's front edge and 112 mm up, round the 7 mm strip of web between the hole and the edge; the spare eye keeps the brake carabiner alone (the "to confirm" item 8 of TKB-DEC-001 is closed). BOM line 14 goes from USD 3 to USD 5	S27 within (7.0 mm wire in a 7.8 mm opening); S28 within (no interference in `cad/fidelity.json`); the fixed end is 6 links long, so the adjustable tail on a 450 mm trunk grows from 10 to 15 links

| Cleat web, all the pull on one slot (TRL 3, after the independent review) | The two legs of the chain need not share its pull: friction round the wrap (a capstan ratio of about 6) lets the split follow how the chain was pulled snug. With all 11.1 kN on one slot at the R1 load, the 35 mm web beside the slot reached 162 MPa, factor 0.77 on the welded proof strength | D-S2: the spine top is raised 20 mm (185 to 205 mm) so the web above the slot bottoms is 55 mm deep; the cheeks grow to 75 mm high and the keeper legs to 73 mm; the slots, notches, pin and chain line do not move. About 0.05 kg more; no cost change | S9 within: 103 MPa, factor 1.21 in the worst split (2.43 shared equally) |

Neither fix changes the working principle, the pitch or the user-facing scope. The strength margin of S30 is not fixed here: it is a safety trade-off (strength against carried mass) and went to Amish as TKB-DEC-001 O7 (decided 2026-10-25: proof load 3.5 kN and a test to failure, option C). It is recorded in TKB-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse it.

Fields not relevant

None. Every field is touched: physics (statics of the self-locking couple, strength of the frame, chain and rope, energy and heat of the lowering, bark contact), chemistry (corrosion of the bare aluminium frame, ageing of polymers), biology (bark and stem tissue of the palm, the body of the person lowered, the rescuer's work at height), mathematics (units, orders of magnitude, the numerical method of the lock and the chain path) and engineering (mass budget, fit of bought parts, manufacturing tolerances).

Concept options

- **Option A, self-locking lever collar** (the current design): rows S1 to S32; 19 within known science, 12 plausible, 1 fails with a fix proposed to Amish (S30, strength margin). The lock (S3) is the deciding unknown.
- **Option B, constricting wrap anchor**: within known science on strength and bark pressure (B2, B3). Its holding claim (B1) is plausible on a capstan estimate (about 3.8 times the load with four wraps and an assumed friction of 0.3), but how a friction hitch tightens round a large, wet, tapering trunk under an axial pull is not covered by a cited source; the rigging time (B4) is an estimate. Not dropped.
- **Option C, screwed stem anchor**: the lug is within known science (C3). The screws' hold (C1) rests on timber formulas that do not apply to palm stem tissue; the permanent holes (C2) fail PR-008 by design; driving four screws by hand at height (C4) is estimated at about 8 minutes, against 5 for the whole rig. Not dropped on the STEM check; it scores lowest in TKB-OPT-001.

Independent review

A second reviewer (a separate headless Claude session given only this STEM check, the calculation note TKB-CAL-001, the product requirements TKB-PRS-001 and `sizing.py`, and none of the author's reasoning) was asked on 2026-10-25 to refute the headline results and every "Within known science" verdict. It recomputed most rows by hand. What it found and what changed:

Table 3. Findings of the independent review.

Row	Finding	Change
S1, S3, S4, S7, S8, S12, S14 to S16, S27 to S29	Upheld (S4: the simplified lock is on the safe side; S8 wording of the shear planes; S12 two strength bases; S14 the descender body near 79 C; S16 1.1 m of loose tail on a 200 mm trunk)	Wording corrected; bases and temperatures stated
S2	Weakened: 30 mm of slack gives 2.6 to 3.0 kN, above the 2.5 kN of R1; 100 mm gives 4.3 kN, beyond the frame; the 30 mm is a drill limit	Plausible; a higher TRL 4 proof load is part of O7
S5	Weakened: pad shear and chain tilt add 4 to 8 mm, so 14 to 18 mm against 20	Plausible
S6	Weakened: the bar also bends vertically under the pad's friction and twists; combined 92 MPa, factor 1.36	Combined check added to the CAL ([C1b]); still within
S9	Refuted for an unequal split: all the pull on one slot gave 162 MPa (factor 0.77)	Fixed under the standing instruction: 55 mm web (D-S2), 103 MPa, factor 1.21 ([C4b])
S11	Weakened: one leg can carry it all; factor 4.1	Restated with one leg
S13	Upheld for a certified device on dry rope; the capstan estimate reaches 150 N at a friction of 0.15	Sensitivity stated
S18	Upheld; 4.55 and 4.62 kg disagreed	Reconciled: the CAL now uses the cited 600 g descender; 4.67 kg with D-S2
S19, S21	Weakened and refuted as worded: green wood is 40 to 50 % weaker and a cocked pad bears at its lower edge, so the margin may be gone; any mark on a palm is permanent	Both plausible; S21 reworded to "no penetration, no mark deeper than 5 mm"
S20	Upheld as a static indentation	Marked static, with the green-wood depth
S22, S23	Weakened: two stable hanging states; the time rests on S17 and an untimed climb	Both plausible
S24	Weakened: a victim to one side adds a sideways pull not in the lock analysis	Stated; within for the vertical sign
S30	Weakened: with the combined bar stress the breaking load is about 5.0 kN (5.1:1), and about 4.5:1 with all of the pull on one slot	Was "Fails, fix proposed"; the fix is a safety trade-off and went to Amish (O7); decided 2026-10-25: test to failure first, now plausible
Missing	A leaning trunk (factor about 1.0 at 30 deg on the adverse side); tether-hole tear-out; asymmetric chain legs; two-person and	S31 (lean, plausible; O8 decided 2026-10-25: 20 deg limit and drill rule) and S32 (tether hole, within) added; the asymmetric case is in S9 and S11; a two-person load is outside the one-person rating (TKB-DDR-001, D7); creep,

Row	Finding	Change
	dynamic loads; pad creep; trunk taper, ridges and algae; wet rope friction	trunk surface and wet rope go to the TRL 4 tests (TKB-RSK-001)
Documents	The review section was still marked pending; dates of sources and documents disagreed	This section; documents dated 2026-10-25

The reviewer would accept the counts only with S2, S5, S19, S21, S22 and S23 downgraded, which is what this version does.

Sources

- Broschat, T. K. "Palm Morphology and Anatomy." ENH1212 (EP473). Gainesville: University of Florida IFAS Extension, 2019. https://edis.ifas.ufl.edu/pdf/EP/EP47300.pdf (checked 2026-10-25). Stem wounds are permanent in palms (S21, C2).
- Rana, M., A. Das, and M. Ashaduzzaman. "Physical and Mechanical Properties of Coconut Palm (*Cocos nucifera*) Stem." *Bangladesh Journal of Scientific and Industrial Research* 50, no. 1 (2015): 39 to 46. https://banglajol.info/index.php/BJSIR/article/view/23808 (checked 2026-10-25). Compressive strength across the grain 9.28 MPa air dry (S19, S20, B3).
- FAO. "Coconut Wood: Processing and Use." Chapter on basic density by zone (100 to 900 kg/m3). https://www.fao.org/4/w7731e/w7731e07.htm (checked 2026-10-25).
- S3i. "Lifting Chain, Grade 80, Alloy Steel." 6 mm: pitch 18 mm, inner width at least 7.8 mm, working load limit 1.12 t, minimum breaking force 45.2 kN, 0.8 kg/m (BS EN 818-2). https://www.s3i.co.uk/lifting-chain-grade-80.php (checked 2026-10-25) (S11, S27).
- Beam Dimensions. "6082 Alloy Aluminium Properties, EN 1999 (welded)." f0 125 MPa and fu 185 MPa for welded 6082-T6. https://beamdimensions.com/materials/Aluminium/EN_1999_2007_Welded/6082 (checked 2026-10-25) (S6, S8, S9).
- Vandernet. "EN 1891 Standard Explained." Type A: at least 22 kN; 15 kN for 3 minutes with figure-eight terminations. https://vandernet.com/tuotetiedostot/EK-FPPFM20x/EN_1891_standard_explained.pdf (checked 2026-10-25) (S12).
- Vandernet. "EN 795 Standard Explained." Static tests at 12 kN (metallic) and 18 kN (textile without a service-life marking). https://www.vandernet.com/tuotetiedostot/EK-FPPFM30x/EN_795_standard_explained.pdf (checked 2026-10-25) (B2).
- SATRA. "EN 341:2011 Descender Devices for Rescue." Classes by descent energy (class A 7.5 MJ), holding force at most 200 N, touch surfaces at most 48 C. https://www.satra.com/spotlight/article.php?id=439 (checked 2026-10-25) (S13, S14, S29).
- Haness Supply. "Petzl D020AA00 I'D S." 600 g, 10 to 11.5 mm rope, 150 kg one person, EN 341 type 2 class A. https://www.hanessupply.com/petzl-d020aa00 (checked 2026-10-25) (S18).
- Piatt, Lance. "Rope Rescue Safety Factors." Rigging Lab Academy, July 3, 2023. https://rigginglabacademy.com/rope-rescue-safety-factors/ (checked 2026-10-25). Teams aim for 5:1, 7:1 or 10:1; 15:1 was once common (S30).
- EN 1995-1-1 (Eurocode 5), design of timber structures: the embedment and yield-moment forms used for the option C estimate are cited by number only; their text was not consulted, and they do not apply to palm stem tissue (C1).

- TKB-CAL-001 (`docs/04-calcs/01-sizing.md`) for every CAL tag in the table.